Preface

Who is This Book For?

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Why Write Yet Another Book About Linear Algebra?

The direct answer to this question is simple: I really think most resources are doing a *bad job* of explaining fundamental linear algebra to new students [1]. Not because they lack material or the complete picture - but because the common approach to teaching fundamental mathematical topics just doesn't work well for most people [2].

Introduction courses to fundamental mathematical topics usually look like this: here'e are some basic definitions, let's derive some specific results from them - now go apply these in your field.

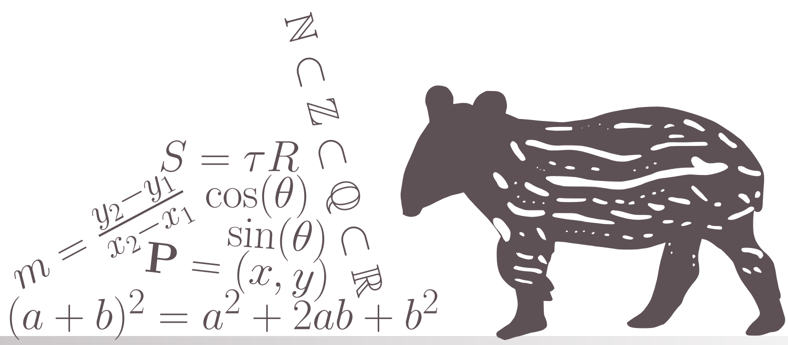
[1] and honestly, this isn't restricted to linear algebra but applies to all fundamental mathematical topics - but it's a classic case

[2] the main exception is students of pure mathematics and people who can easily deal with extremely abstract concepts - a very specific group of people at the end of the day

CHAPTER

1

THE BASICS OF LINEAR ALGEBRA



1.1 VECTORS AND VECTOR SPACES

1.1.1 MOTIVATION

Motivating Problem 1.1 Gravity Force

Consider a simple simulation of classical ("Newtionian") gravity, say of a planet with mass m orbiting a star with mass M (where $M \gg m$). At any given moment, the force acting on the planet due to the star's gravity has magnitude

$$F = G\frac{Mm}{r^2}, \tag{1.1}$$

where r is the distance between the planet and the star at that moment, and G is a universal constant (specifically, $G = 6.674 \times 10^{-11}\,\mathrm{N\,m^2\,kg^{-2}}$).

However, since we live in 3-dimensional space[a], we also want this force to be pointing in the direction from the planet to the star. To do this, we need to understant how to describe orientations in 3-dimensional space, and be able to pack together both orientation and magnitude in the correct way.

[a] allegedly.

Motivating Problem 1.2 Bouncing/Reflecting Off a Wall

Imagine we are writing a 3D computer game in which some object is bouncing off a wall. Alternatively, consider 3D graphics in general, where one simulates rays of light reflecting off of smooth surfaces. In both cases the same problem needs to be solved: given a flat plane of some orientation in space, and an object hitting it from some direction - what would be the direction of the object after it bounces off the plane?

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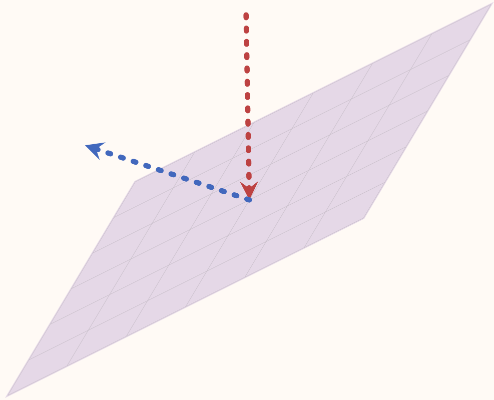

To solve this problem we will need to know how to represent quantities like position, velocity and orientation in 3D, and more generally points, lines and planes - and to know how to calculate the intersection between these different objects. By the end of this section you should be able to solve this problem by yourself, and even generalize it to any number of dimensions.

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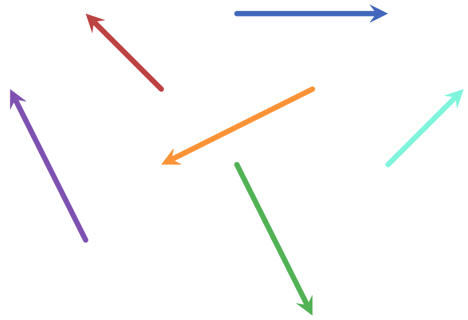
1.1.2 VECOTRS AND THEIR GEOMTERIC PROPERTIES

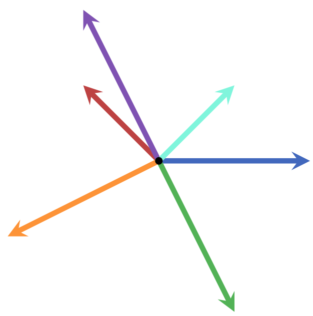
The above motivating problems require objects that have both a magnitude and a direction: for example, the force of gravity between a star and a planet has magnitude given by Equation 1.1 and a direction from the planet to the star. In Motivating-problem 1.2, if we model some object hitting a wall, the object's velocity is a combination of magnitude (which we usually simply call speed) and a direction (towards the wall).

In the simplest terms, vectors are mathematical objects which have both a magnitude and a direction. The magnitude of a vector is called norm, while its direction is sometimes refered to as its orientation. Vecotrs can exist in 2-dimesnsions, 3-dimensions or even higher dimensions. Since higher dimensions are rather hard to conseptulize, I will first focus on 2- and 3-dimensional vectors so that we can better undetstand their properties, and only then generalize to higher dimensions when it's relevant.

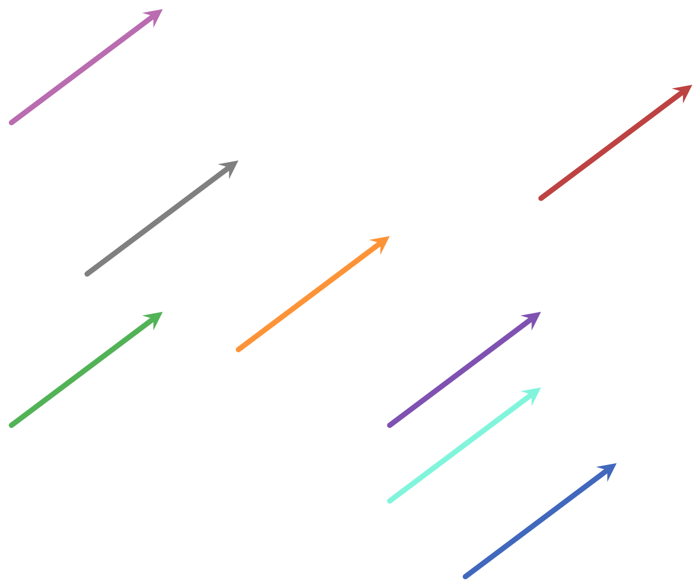
In 2-dimensions we can represent vectors as arrows:



However, since we are usually interested in the magnitude and direction of vectors and not where they originate from, we draw them as originating in the same location which we unimaginatively call the origin:



In fact, all vectors that have the same norm and orientation are considered identical - no matter where they originate. For example, the following vectors are actually all identical:



Since we will be using vectors as mathematical objects and would like to use them in an arithmetical context, we need to use some kind of notation. Usually, vectors are denoted as lower-case Latin letters, either in boldface or with a line or an arrow symbol above or below the letter. Here are few examples:

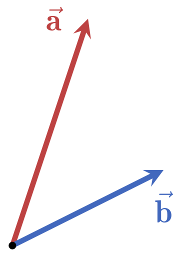
$$\mathbf{a},\ \mathbf{b},\ \mathbf{c},\ \mathbf{x},\ \mathbf{y},\ \mathbf{z},\ \ldots$$
$$\overline{a},\ \overline{b},\ \overline{c},\ \overline{x},\ \overline{y},\ \overline{z},\ \ldots$$
$$\underline{a},\ \underline{b},\ \underline{c},\ \underline{x},\ \underline{y},\ \underline{z},\ \ldots$$
$$\vec{a},\ \vec{b},\ \vec{c},\ \vec{x},\ \vec{y},\ \vec{z},\ \ldots$$
$$\vec{\mathbf{a}},\ \vec{\mathbf{b}},\ \vec{\mathbf{c}},\ \vec{\mathbf{x}},\ \vec{\mathbf{y}},\ \vec{\mathbf{z}},\ \ldots$$

In this book I am using the last notation - a boldface Latin letter with an arrow above it. This ensures that it is both easy to read and is distinctive enough to quickly differentiate vectors from other objects. Sometimes, depending on context, the vectors are colored to make it easier to follow through (the process of setting up/developing a concept/equation).

There are two fundamental operations involving vectors: addition and scaling. These two operations result in creating a new vector of the same dimensions: adding together two 2-dimensional vectors results in a 2-dimensional vector, scaling a 3-dimensional vector result in a 3-dimensional vector, etc.

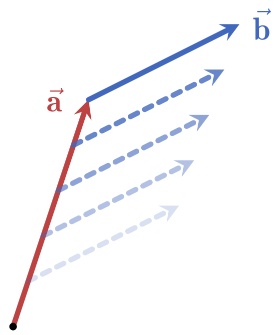
Let's have a closer look at these two fundamental vector operations:

- **Adding vectors together**: any two vectors of the same dimensionality (e.g. 2-dimensional, 3-dimensional, 4-dimensional, etc.) can be added together, and the result is a third vector. Pictorially, here are two vectors $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ which we want to sum together:

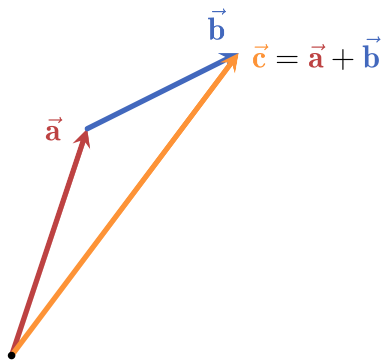


The steps to produce the sum of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$ is the following:

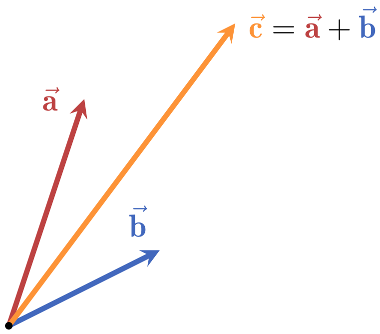
1. move $\vec{\mathbf{b}}$ such that its origin aligns with the head of $\vec{\mathbf{a}}$:



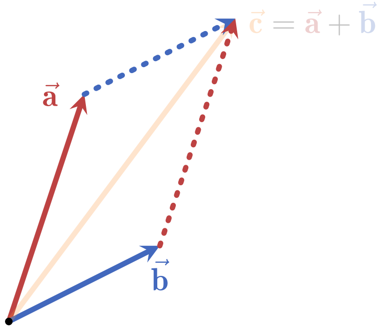
2. draw a new vector from the origin to the head of the shifted $\vec{\mathbf{b}}$:



3. the resulting vector $\vec{\mathbf{c}}$ *is* the sum of $\vec{\mathbf{a}}$ and $\vec{\mathbf{b}}$. Here are all three vectors together from the same origin:

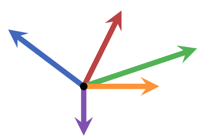


This method is called the triangle method, and it displays an important property of vector addition: it is commutative! This means that the order of addition doesn't matter: $\vec{\mathbf{a}} + \vec{\mathbf{b}} = \vec{\mathbf{b}} + \vec{\mathbf{a}}$. To see this visually, we can apply the triangle method twice, i.e. moving both $\vec{\mathbf{a}}$ *and* $\vec{\mathbf{b}}$:

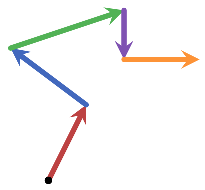


We see that no matter which vector we add onto which (i.e. what is the order of addition) - the result is the same. The shape formed by this demonstration is called a parallelogram, and it is incredibly important in linear algebra as a whole as we will see in future sections.

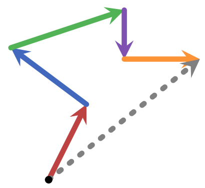
This method of summing vectors can be applied to any number of vectors. For example, consider the following 5 vectors:



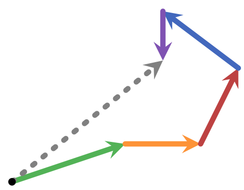
To sum them together, all we need is to start with one the vectors (say, the red one) and use the triangle method to add the other vectors step by step:



Finally, we connect the origin of the first vector with the head of the last vector (in this case the orange one):

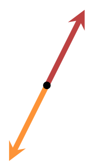


and that's it, the dashed gray vector is the sum of all five vectors. To see that the order of addition really doesn't matter we can do this again but this time starting with the green vector and changing the summation order completely:



As can be seen, the resulting dashed gray vector is exactly the same as before.

Another important property of the addition of vectors is that we can always take a vector and rotate it by 180°, giving us a vector that "cancels" the original vector by summation. For example, the following red vector is rotated to yield its opposite, colored in orange:



These vectors added together using the triangle method yield a vector with 0 norm:

(here I drew the vectors a bit shifted from each other so that we can see both)

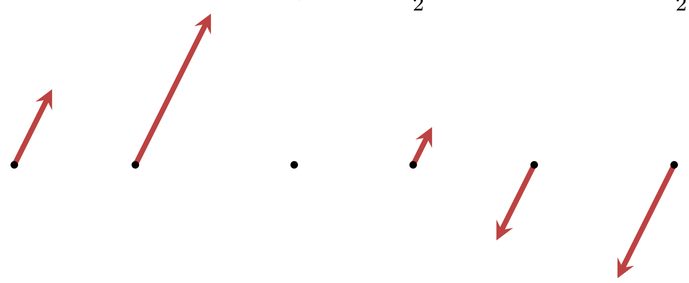
Such a vector is called the zero vector, denoted as $\vec{\mathbf{0}}$. It has no direction, and a norm of 0. Adding $\vec{\mathbf{0}}$ to any other vector is the same as not adding any vector at all (and it doesn't matter "from which side" we add $\vec{\mathbf{0}}$):

$$\vec{\mathbf{a}} + \vec{\mathbf{0}} = \vec{\mathbf{0}} + \vec{\mathbf{a}} = \vec{\mathbf{a}}. \tag{1.2}$$

We say that the zero vector is *agnostic* (also: neutral) to addition.

- **Scaling vectors**: any vector can be scaled by any real number, which in this context we call a scalar. Scaling by a number simply "stretches" the vector by the number. For example, in the following figure a red vector is scaled by a scalar λ, taking on different values:

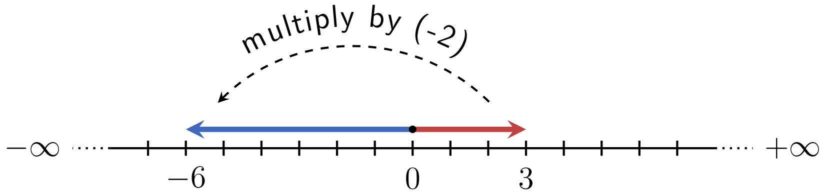
$\lambda = \quad 1 \quad 2 \quad 0 \quad \frac{1}{2} \quad -1 \quad -\frac{3}{2}$



This example shows us several things:

- no matter by which positive scalar we scale a vector, it always points in the same direction.
- If we scale a vector by 0 it becomes the zero vector.
- Finally, scaling a vector by a negative scalar rotates the vector by 180° and then scales it by the absolute value of the scalar.

This is similar to how real numbers behave under multiplication: multiplying a real number by a negative number is, in a sense, "flipping" and scaling it relative to 0 (the origin). For example, multiplying 3 by -2 flips the the number to point to the left, and then scales it to be of length 6:



Therefore, scaling a vector is denoted just like multiplication of real numbers: $\lambda\vec{\mathbf{a}}$, where λ is the scalar. In the above example, if the red vector is $\vec{\mathbf{a}}$, then the other five vectors can be written as $2\vec{\mathbf{a}}$, $0\vec{\mathbf{a}}$, $\frac{1}{2}\vec{\mathbf{a}}$, $-1\vec{\mathbf{a}}$ and $-\frac{3}{2}\vec{\mathbf{a}}$.

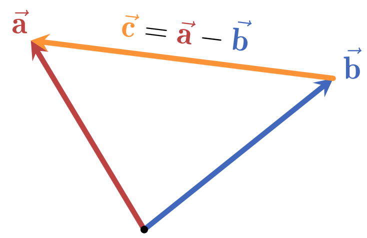
Furthermore, we can place the vector notation in different places and even hiding the scalar depending on convinience, so for example we can write $-\vec{\mathbf{a}}$ instead of $-1\vec{\mathbf{a}}$ and $-\frac{3\vec{\mathbf{a}}}{2}$ instead of $-\frac{3}{2}\vec{\mathbf{a}}$.

Just like with real numbers, we call $-\vec{\mathbf{a}}$ the additive inverse of $\vec{\mathbf{a}}$.

In fact, the similarities to real number don't end there: subtracting a vector $\vec{\mathbf{b}}$ from another vector $\vec{\mathbf{a}}$ is simply adding $\vec{\mathbf{a}}$ and the additive inverse of $\vec{\mathbf{b}}$:

$$\vec{\mathbf{a}} - \vec{\mathbf{b}} = \vec{\mathbf{a}} + \left(-\vec{\mathbf{b}}\right). \tag{1.3}$$

Visually, if we place the origin of $\vec{\mathbf{a}} - \vec{\mathbf{b}}$ at the head of $\vec{\mathbf{b}}$, its head would be at the same point as the head of $\vec{\mathbf{a}}$:



While subtration is not commutative, it is antisymmetric:

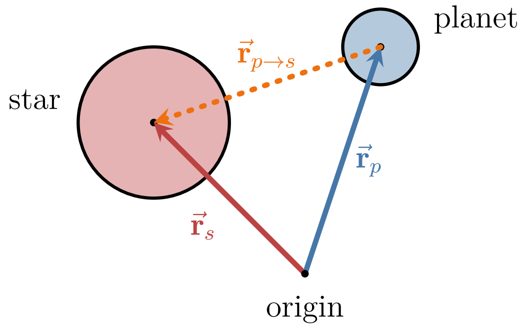
$$\vec{\mathbf{a}} - \vec{\mathbf{b}} = -\left(-\vec{\mathbf{a}} + \vec{\mathbf{b}}\right) = -\left(\vec{\mathbf{b}} + \left(-\vec{\mathbf{a}}\right)\right) = -\left(\vec{\mathbf{b}} - \vec{\mathbf{a}}\right), \tag{1.4}$$

i.e. flipping the order of subtraction changes the sign of the expression.

So we can use vector subtraction as a convinient way to point from one vector to another. This can be utilized to implement the gravity force in Motivating-problem 1.1: if we know the position of the planet (let's call it $\vec{\mathbf{r}}_p$) and the position of the star ($\vec{\mathbf{r}}_s$), a vector pointing from the planet's center to the star's center would be, for example,

$$\vec{\mathbf{r}}_{p\to s} = \vec{\mathbf{r}}_p - \vec{\mathbf{r}}_s. \tag{1.5}$$

(the notation $p \to s$ is just an easy way to represent that the vector is pointing from the planet to the star)



However, this vector has a norm equal to the distance between the plant and the star - and we want to use it to calculate the *force* acting on the planet, which has a different magnitude (and units). To do this, we need to be able to scale $\vec{\mathbf{r}}_{p\to s}$ to whatever norm we want. Taking inspiration from real numbers, we can employ the following procedure: say we want to scale some non-zero number x to be y. We can multiple x by $\frac{y}{x}$, canceling the x:

$$x \mapsto \cancel{x} \cdot \frac{y}{\cancel{x}} = y.$$

A different way to phrase this is that we simply multiplying (or *scaling*) x by $\frac{1}{x}$, and then multiply the result by y:

$$x \mapsto \cancel{x} \cdot \frac{1}{\cancel{x}} \cdot y = y.$$

We can apply the same technique to vectors. Scaling a vector by its *reciprocal norm* yields a vector with norm 1, which we denote by replacing the arrow in the vector notation by a "hat":

$$\hat{\mathbf{a}} = \frac{1}{\|\vec{\mathbf{a}}\|}\vec{\mathbf{a}}. \tag{1.6}$$

This operation is called normalization, and the resulting vector is a unit vector (also: normalized vector).

Now, if we scale a unit vector $\hat{\mathbf{a}}$ by any scalar λ we get a vector of the same direction as $\hat{\mathbf{a}}$ (and also $\vec{\mathbf{a}}$), and with a norm of λ. The vectors $\vec{\mathbf{a}}$, $\hat{\mathbf{a}}$ and $\lambda\hat{\mathbf{a}}$ are all parallel to each other.

Back to our star-planet system: by normalizing the vector $\vec{\mathbf{r}}_{p\to s}$ we get a unit vector pointing from the planet to the star: $\hat{\mathbf{r}}_{p\to s}$, and scaling by the magnitude of the gravitational attraction (Equation 1.1) we finally get the correct *gravity force vector*:

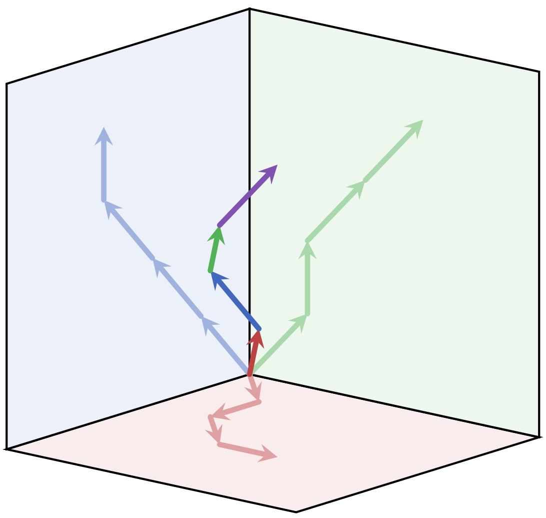
$$\underbrace{\boxed{\vec{\mathbf{F}}}}_{\text{vector}} = \underbrace{\boxed{G\frac{Mm}{r^2}}}_{\text{scalar}}\underbrace{\boxed{\hat{\mathbf{r}}_{p\to s}}}_{\text{vector}} \tag{1.7}$$

Indeed, $\vec{\mathbf{F}}$ is a vector: it has a magnitude (the scalar $G\frac{Mm}{r^2}$) and a direction (the unit vector $\hat{r}_{p\to s}$). Any vector can be written as a unit vector scaled by a real number:

$$\vec{\mathbf{a}} = a\hat{\mathbf{a}}, \tag{1.8}$$

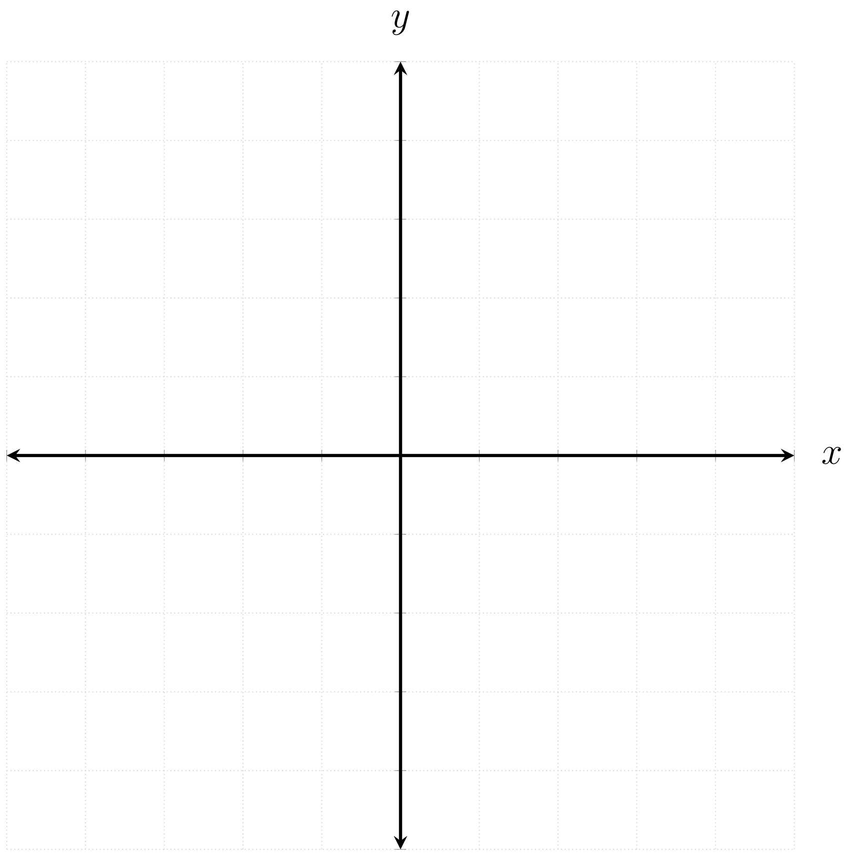
where $a \in \mathbb{R}$ is the magnitude of $\vec{\mathbf{a}}$, i.e. $a = \|\vec{\mathbf{a}}\|$. This is nothing more than a rearrangement of Equation 1.6: by dividing both sides of the equation by $a = \|\vec{\mathbf{a}}\|$ we get the definition of a unit vector.

TBW: text about how in 3D the same rules apply.



1.1.3 CARTESIAN COORDINATE SYSTEM AND BASIS SETS

Everything we discussed so far is very nice - but it doesn't give us much in the way of actually doing any practical calculations. To do that, we need to "quantify" vectors somehow. By far the most common way in 2- and 3-dimensions is to use a Cartesian coordinate system: in dimensions we simply choose a direction to be the x-direction (usually the horizontal direction such that the numbers increase to the right), and then the choose between the two possible perpendicular directions to be the positive y-direction:



We then set the origin of all vectors to be at the origin of the axes (i.e. the point $(0, 0)$). To quantify a vector, we draw two lines from its head: one parallel to the x-axis and one parallel to the y-axis. The points at which these lines intersect the y- and x-axes are then, respectively, the x- and y-coordinates of the vector: for example, Figure 1.1 shows a vector $\vec{\mathbf{a}}$ placed in a Cartesian coordinate system. The x-coordinate of the vector is $a_x = 2$ and its y-coordinate is $a_y = 3$.

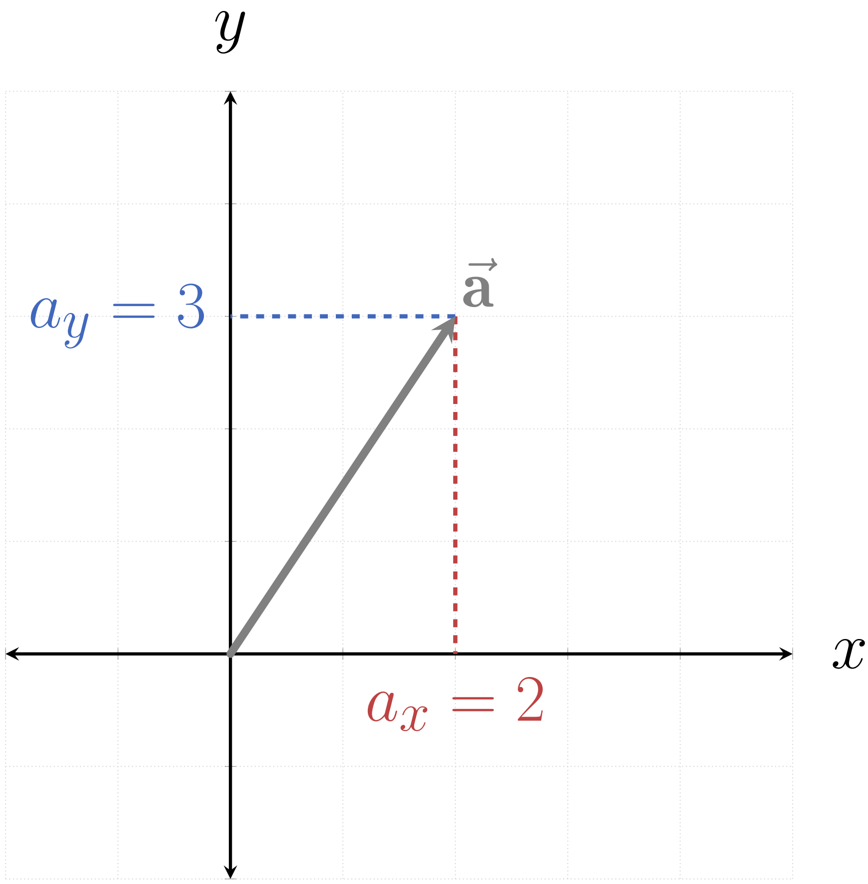


Figure 1.1: The vector $\vec{\mathbf{a}}$ placed in a Cartesian coordinate system, making its coordinates $a_x = 3$ and $a_y = 2$.

Using these coordinates, we can then write $\vec{\mathbf{a}} = (3,2)$, i.e. identify the vector with the point in the Cartesian system where its head is (given that its origin aligns with the origin of the system). In general, this works for any vector: each point $\mathbf{P} = (x,y)$ in the Cartesian system can be identified with a single vector $\vec{\mathbf{v}} = (x,y)$, and each vector can be identified with a single point.

However, vector are not points, and we should make this fact reflected in the notation we use. Therefore I choose to use the column-form notation for vectors, for example

$$\vec{\mathbf{a}} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \ \vec{\mathbf{v}} = \begin{bmatrix} x \\ y \end{bmatrix}, \tag{1.9}$$

etc. In principle, I could have chosen to use a similar notation, namely the row-form notation:

$$\vec{\mathbf{a}} = \begin{bmatrix} 2 & 3 \end{bmatrix}, \ \vec{\mathbf{v}} = \begin{bmatrix} x & y \end{bmatrix}. \tag{1.10}$$

While both notations are essentially equivalent, to keep notational consistency I would need to strictly use one of these notations, and for reasons of personal preferences I chose the column-form. This enforces the use of some notational conventions later, for example the meaning of matrix columns vs. rows and the notation for a set of related objects called covectors (discussed in details in REF).

How does the column form of a vector behaves under the two basic vector operations (addition and scaling)? Let's see what happenes when we add two vectors together, say

$$\vec{\mathbf{a}} = \begin{bmatrix} 1 \\ 3 \end{bmatrix}, \quad \vec{\mathbf{b}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Using the triangle method, we move $\vec{\mathbf{b}}$ such that its origin lies at the head of $\vec{\mathbf{a}}$ (Figure 1.2). The result of the addition $\vec{\mathbf{c}} = \vec{\mathbf{a}} + \vec{\mathbf{b}}$ ...

Another way of interpreting the coordinates of a vector is to define two unit vectors:

$$\hat{x} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \hat{y} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{1.11}$$

Looking at Figure 1.3, these two vectors correspond to the two axes of the Cartesian coordinate system: $\hat{x}$ is a unit vector in the direction of the x-axis, and $\hat{y}$ is a unit vector in the direction of the y-axis.

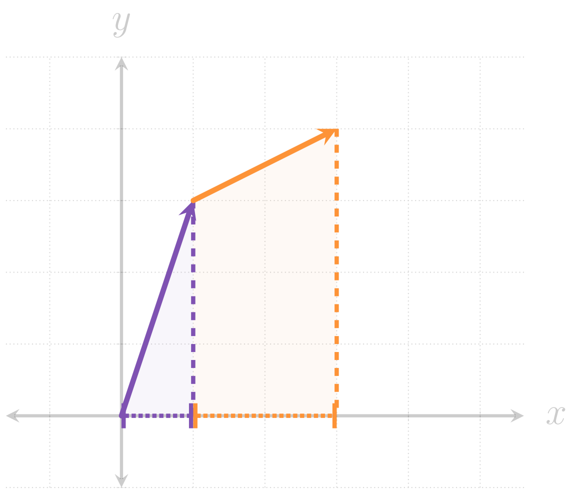


Figure 1.2: Text text text.

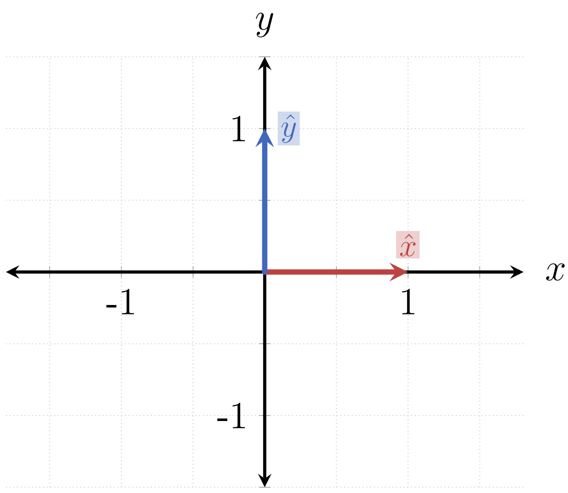


Figure 1.3

1.1.4 GEOMTERIC OBJECTS

1.1.5 PROJECTIONS AN'D REJECTIONS

1.1.6 VECTOR SPACES

1.1.7 SOLVING THE MOTIVATING PROBLEMS

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